

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title: Quiz Application

1. Introduction

1.1 Purpose

The purpose of this document is to describe the software requirements of the Quiz Application. It defines the features, functions, and technical requirements necessary to develop the application.

1.2 Scope

The Quiz Application is a frontend-based website that enables users to participate in quizzes containing multiple-choice questions. It includes a timer, randomized questions, automatic score calculation, and a scoreboard.

The application is designed to be simple, interactive, and accessible through a web browser.

1.3 Intended Users

- Students who want to test their knowledge.
- Teachers who want to demonstrate online quizzes.
- Users who want to practice MCQs.

2. Overall Description

2.1 Product Perspective

The Quiz Application is a standalone frontend web application developed using HTML, CSS, and JavaScript. It does not require a backend server for its basic functionality.

2.2 Product Functions

The main functions of the application are:

- Starting a quiz.
- Displaying MCQs and answer options.
- Allowing users to select answers.
- Randomizing the order of questions.
- Managing the countdown timer.
- Checking answers and calculating scores.
- Displaying the final result.

2.3 User Characteristics

The application is intended for users with basic computer and web browsing skills. No specialized technical knowledge is required.

2.4 Operating Environment

- **Operating System:** Windows, Linux, or macOS.
- **Frontend Technologies:** HTML, CSS, JavaScript.
- **Browser:** Google Chrome, Microsoft Edge, Mozilla Firefox, or another modern web browser.
- **Hardware:** A computer or laptop capable of running a modern web browser.

3. Functional Requirements

FR-1: Start Quiz

The system shall allow the user to start the quiz using a start button.

FR-2: Display Questions

The system shall display one multiple-choice question at a time, along with the available answer options.

FR-3: Answer Selection

The system shall allow the user to select an answer from the available options.

FR-4: Question Randomization

The system shall display questions in a randomized order when a quiz attempt begins.

FR-5: Timer Management

The system shall display a countdown timer and track the remaining quiz time.

FR-6: Automatic Submission

The system shall end the quiz automatically when the timer reaches zero.

FR-7: Score Calculation

The system shall check the selected answers against the correct answers and calculate the final score.

FR-8: Scoreboard

The system shall display the final score after the quiz is completed.

FR-9: Quiz Completion

The system shall allow the user to finish the quiz and view the result.

4. Non-Functional Requirements

4.1 Usability

The interface should be simple, clear, and easy to understand.

4.2 Performance

Questions, answer options, and results should appear without unnecessary delays.

4.3 Reliability

The application should calculate scores correctly and handle timer completion properly.

4.4 Compatibility

The application should work on modern web browsers.

4.5 Maintainability

The HTML, CSS, and JavaScript code should be organized so that future modifications can be made easily.

4.6 Responsiveness

The interface should adapt to different screen sizes, including laptops, tablets, and mobile phones, if responsive styling is implemented.

5. Hardware and Software Requirements

Hardware Requirements

- Computer or laptop.
- Keyboard and mouse or touchpad.
- Minimum 4 GB RAM recommended.

Software Requirements

- Windows, Linux, or macOS.
- Visual Studio Code or another code editor.
- HTML, CSS, and JavaScript.
- A modern web browser.

6. System Workflow

1. The user opens the Quiz Application.

2. The user clicks the Start Quiz button.
3. The system randomizes and displays the questions.
4. The timer starts counting down.
5. The user selects answers.
6. The system tracks the selected answers.
7. The quiz ends when all questions are completed or the timer expires.
8. The system calculates the score.
9. The scoreboard displays the final result.

7. Constraints and Limitations

- The basic application works on the frontend.
- User accounts and login functionality are not included in the basic version.
- Scores are not permanently stored unless a storage mechanism is implemented.
- The application requires a compatible web browser.
- Question randomization changes the order of questions; randomizing answer options would require an additional feature.

8. Future Enhancements

- User registration and login.
- Multiple quiz categories.
- Different difficulty levels.
- Database integration for storing questions and scores.
- Leaderboard functionality.
- Performance history and analytics.

9. Conclusion

The SRS defines the functional and non-functional requirements of the Quiz Application. It provides a clear foundation for developing a user-friendly frontend application with MCQs, a countdown timer, randomized questions, automatic score calculation, and a scoreboard.